Neuron Games for Spiking Networks: A Constrained State-Based Potential Markov Game with Kernel/DPP Diversity, Metabolic Barriers, and Eligibility-Shapley Explainability

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Abstract

We formulate the training of spiking neural networks (SNNs) as a multi-agent game in which each neuron is a player that adapts its incoming synapses. A central nuance—often missed by "neuron competition" metaphors—is *induced-feature dependence*: when a presynaptic neuron changes its synapses, it changes a postsynaptic neuron's effective response map even if the postsynaptic neuron does nothing. We capture this effect with a state-based potential Markov game. The state includes synapses, membrane trajectories, and synapse-local eligibility traces, and the game's reward is the improvement of a global potential across adaptation windows. Our primary learning rule is three-factor plasticity—eligibility trace times a neuromodulatory learning signal—grounded in neuromodulated spike-timing-dependent plasticity theory [3].

Two mechanisms make the game stable and non-degenerate. First, a diversity reward is built from point-process or spike-train kernels and, optionally, a determinantal point process (DPP) log-determinant objective; this avoids correlation metrics that are poorly suited to sparse temporal spikes [5, 8]. Second, a homeostatic energy budget with barrier terms prevents max-rate "shouting matches" and yields a constrained Markov potential game viewpoint [12]. Finally, we connect training and interpretability through *Eligibility/Temporal Shapley*: Shapley-style marginal contributions are computed over plasticity traces rather than over ablated neuron subsets, avoiding infeasible subset evaluation [9]. Backpropagation appears only as a non-biological limiting case involving smooth rates and exact error broadcasts, consistent with eligibility-propagation perspectives [4].

1 Introduction

Backpropagation remains the dominant training paradigm, yet it is widely viewed as biologically implausible because it relies on non-local gradient transport and precise, layer-specific credit assignment. Biological learning, by contrast, is often modeled through local pre–post coincidence modulated by a third factor, such as reward, neuromodulation, or a teaching signal. Eligibility traces bridge the temporal delay between local activity and that modulatory signal [3].

Many game-theoretic neuron models treat diversity and stability as afterthoughts. If neurons are selfish and optimize only immediate payoff, a layer can collapse into redundant solutions in which all neurons learn the same easy feature. Firing rates can also diverge as every neuron attempts to dominate the learning signal. Our position is that diversity and homeostasis must be mechanisms that make the game function, not decorative penalties.

1.1 Induced-feature dependence

A neuron's feature is not a static label. It is an input-to-spike mapping induced by upstream activity. When upstream neurons change their synapses, they alter the downstream neuron's

input spike trains, membrane dynamics, and spike responses—even before the downstream synapses update. A game-theoretic formulation must therefore be state-based: strategies act on synapses, while features emerge from an evolving state.

1.2 Contributions

The paper makes five contributions:

1. We formulate SNN learning as a *state-based potential Markov game*: neurons are players; the state contains synapses, membrane variables, spike trains, and eligibility traces; and reward is potential improvement [2].

2. We define point-process kernel diversity and DPP log-determinant diversity on spike trains or eligibility traces, avoiding correlation measures that fail for sparse temporal events [5, 8].

3. We combine metabolic and barrier-based homeostasis. The resulting energy model constrains the potential game and prevents degenerate equilibria [12].

4. We introduce Eligibility-Shapley explainability and modulation: Shapley-style credit is computed over plasticity traces, providing training-time attribution without exponential subset evaluation [9, 10].

5. We identify backpropagation as a limiting case. Under smooth-rate dynamics and exact error broadcasts, the three-factor rule reduces to gradient descent, consistent with eligibility-propagation views [4].

2 Related Work

Potential games and state-based potential games. Potential games align individual improvements with a global potential [1]. State-based potential games extend this idea to systems in which an evolving state coordinates agent behavior [2].

Three-factor plasticity and eligibility traces. Eligibility traces and neuromodulated STDP provide a canonical biology-forward learning template [3].

Eligibility propagation. E-prop decomposes learning into synapse-local eligibility traces and broadcast learning signals, achieving strong performance on temporal tasks without full backpropagation through time (BPTT) [4].

Spike-train kernels and MMD for point processes. Kernel methods define positive definite similarities for spike trains and thereby enable reproducing-kernel Hilbert space (RKHS) metrics [5]. Spike-train kernels combined with maximum mean discrepancy (MMD) can compare spike-train distributions directly [6, 7].

DPP diversity in neural networks. DPP-based regularizers provide principled log-determinant diversity objectives for neural representations [8].

Shapley explainability. SHAP applies Shapley values to machine-learning feature attribution [9]. Neuron Shapley estimates Shapley values for neurons or filters and demonstrates neuron-level faithfulness properties [10]. Shapley–Taylor indices extend the framework to interaction attributions up to a selected order [11].

Constrained Markov potential games. Recent work studies constrained Markov potential games and independent learning dynamics within them, providing a natural mathematical lens once energy budgets and barriers are introduced [12].

3 Preliminaries

3.1 Spiking notation

Neuron i emits the spike train

$$s_i(t) = \sum_k \delta(t - t_i^k). \tag{1}$$

We use leaky integrate-and-fire-style dynamics over each window $[t_n, t_{n+1}]$:

$$\tau_m \dot{u}_i(t) = -u_i(t) + \sum_j w_{ij}(\kappa * s_j)(t) + I_i(t), \tag{2}$$

where threshold and reset dynamics produce the spikes $s_i(t)$.

3.2 Three-factor learning

A three-factor synaptic update takes the form

$$\Delta w_{ij}^{(n)} = \eta \int_{t_n}^{t_{n+1}} m_i(t) e_{ij}(t)\, dt, \tag{3}$$

where $e_{ij}(t)$ is a synapse-local eligibility trace and $m_i(t)$ is a neuromodulatory, possibly broadcast, learning signal [3]. Eligibility traces retain local "would-have-learned" information and thereby bridge temporal delays in behavioral learning.

3.3 Kernels on spike trains and RKHS distances

Spike trains can be embedded in a Hilbert space through a positive definite kernel $k(\tau, \tau')$, enabling distances and divergences appropriate for point processes [5]. MMD is a standard RKHS distance between distributions and admits efficient estimators [6].

3.4 Shapley values and interaction attribution

Shapley values allocate marginal contributions in cooperative games, and SHAP uses these principles for machine-learning explanations [9]. Shapley–Taylor indices extend the idea to interactions of order at most k [11].

4 Neuron Games: A Constrained State-Based Potential Markov Game

4.1 Players and actions

The players are neurons $i \in \{1, \dots, N\}$. At adaptation window n, neuron i selects updates to its incoming synapses:

$$a_i^{(n)} := \left\{ \Delta w_{ij}^{(n)} \right\}_{j \in \mathrm{pre}(i)} \in \mathcal{A}_i. \tag{4}$$

Actions may be continuous, representing graded plasticity, or discrete, representing potentiation and depression choices.

4.2 State and induced-feature dependence

The global state at window n is

$$x^{(n)} = \left(W^{(n)}, U^{(n)}(\cdot), S^{(n)}(\cdot), E^{(n)}(\cdot), Z^{(n)}\right), \tag{5}$$

where W contains synaptic weights; $U(\cdot)$ contains membrane trajectories; $S(\cdot)$ contains spike trains; $E(\cdot)$ contains eligibility traces; and Z contains optional homeostatic variables such as thresholds and running rates.

The effective feature of neuron i during window n is the mapping from the stimulus history to its spike response under state $x^{(n)}$. When an upstream neuron updates, $S_j(\cdot)$ changes, which changes $U_i(\cdot)$ and thus the induced feature of neuron i without neuron i acting. This dependence motivates the state-based Markov formulation.

4.3 Transition kernel

Given state $x^{(n)}$ and joint action $a^{(n)} = (a_1^{(n)}, \ldots, a_N^{(n)})$, the system first updates the synapses,

$$W^{(n+)} = W^{(n)} + \Delta W^{(n)}, \tag{6}$$

and then rolls out the spiking dynamics over $[t_{n+1}, t_{n+2}]$ to obtain $U^{(n+1)}(\cdot)$, $S^{(n+1)}(\cdot)$, $E^{(n+1)}(\cdot)$, and $Z^{(n+1)}$. This defines the Markov transition kernel

$$x^{(n+1)} \sim P\left(\cdot \mid x^{(n)}, a^{(n)}\right). \tag{7}$$

4.4 Potential function

We define the state-based potential

$$\Phi(x) = -\mathcal{J}(x) + \beta\mathcal{V}(x) - \mu\mathcal{E}(x) - \lambda R(W). \tag{8}$$

Here, $\mathcal{J}(x)$ is a task objective estimated over the window, such as supervised loss, a reinforcement-learning return proxy, or predictive-coding energy; $\mathcal{V}(x)$ is a kernel or DPP diversity reward; $\mathcal{E}(x)$ is a metabolic and barrier energy cost; and $R(W)$ is a conventional weight regularizer.

The scalar reward is the potential improvement

$$r^{(n)} = \Phi\left(x^{(n+1)}\right) - \Phi\left(x^{(n)}\right). \tag{9}$$

This follows the state-based potential-game view in which the evolving state coordinates multi-agent adaptation [2].

4.5 Constrained Markov potential game viewpoint

When $\mathcal{E}(x)$ includes barrier terms that diverge near physical or numerical limits, the resulting model can be interpreted through the lens of a constrained Markov potential game (CMPG) [12].

5 Diversity: Point-Process Kernels and DPP Log-Determinants

5.1 Why correlation is insufficient

Spikes are sparse temporal point processes, so correlations between spike counts can be poor proxies for representational overlap. Kernel methods provide positive definite similarities designed for structured temporal objects [5].

5.2 Feature objects: spikes or plasticity signatures

For neuron i in layer ℓ, define a temporal feature object τ_i over a window. Two choices are

$$\tau_i := s_i(\cdot) \quad \text{or} \quad \bar{s}_i(t) = (\kappa * s_i)(t), \qquad \text{(spike-based)}, \tag{10}$$

$$\tau_i := \{e_{ij}(\cdot)\}_{j \in \mathrm{pre}(i)}, \qquad \text{(plasticity-based)}. \tag{11}$$

The second choice is the neuron's *plasticity signature*. Because eligibility traces are already maintained for learning, their reuse for diversity requires little additional state.

5.3 Kernel redundancy penalty

Let $k(\tau_i, \tau_j)$ be a spike-train or trace-train kernel, and let the layerwise Gram matrix K_ℓ have entries $(K_\ell)_{ij} = k(\tau_i, \tau_j)$. For the neuron set $\mathcal{N}_\ell$, define

$$\mathcal{R}_\ell(x) = \sum_{\substack{i,j \in \mathcal{N}_\ell \\ i \neq j}} (K_\ell)_{ij}. \tag{12}$$

The corresponding diversity reward is

$$\mathcal{V}_{\mathrm{kernel}}(x) = -\sum_\ell \mathcal{R}_\ell(x). \tag{13}$$

Highly overlapping plasticity signatures therefore decrease the potential.

5.4 MMD diversity

Treat $\{\tau_i : i \in \mathcal{N}_\ell\}$ as samples from a layerwise representation distribution P_ℓ. Diversity can be encouraged by penalizing concentration or maximizing RKHS dispersion relative to a reference distribution Q_ℓ. MMD supplies a principled distributional distance and scalable estimators [6]; with spike-train kernels, it can operate directly on temporal point-process data [7].

5.5 DPP log-determinant diversity

A stronger within-layer diversity reward is

$$\mathcal{V}_{\mathrm{DPP}}(x) = \sum_\ell \log\det(K_\ell + \varepsilon I), \tag{14}$$

where $\varepsilon > 0$ ensures numerical stability. Maximizing the log-determinant rewards repulsion and diversity and follows DPP-based diversity models for neural networks [8]. Naive evaluation costs $O(|\mathcal{N}_\ell|^3)$ per layer; practical implementations can use layer miniblocks, low-rank approximations, or Nyström methods.

6 Homeostasis: Metabolic Cost and Barriers

6.1 Metabolic energy budget

Define the window-energy functional

$$\begin{aligned}\mathcal{E}_{\mathrm{met}}(x) = c_s \sum_i \int_{t_n}^{t_{n+1}} s_i(t)\, dt + c_u \sum_i \int_{t_n}^{t_{n+1}} u_i(t)^2\, dt \\ + c_w \sum_{ij} |w_{ij}|^p.\end{aligned} \tag{15}$$

The first term charges a cost for each spike and discourages max-rate firing to capture credit. The second penalizes sustained excitation, and the third represents synaptic maintenance. Together, these terms give biologically interpretable incentives and stabilize the multi-agent dynamics.

6.2 Barrier functions

Suppose the physical and numerical constraints are

$$w_{ij} \in (w_{\min}, w_{\max}), \qquad u_i(t) \in (u_{\min}, u_{\max}). \tag{16}$$

Define the log-barrier cost

$$\begin{aligned} \mathcal{E}_{\text{bar}}(x) = & -\sum_{ij} \left[\log(w_{ij} - w_{\min}) + \log(w_{\max} - w_{ij})\right] \\ & -\sum_{i,t} \left[\log(u_i(t) - u_{\min}) + \log(u_{\max} - u_i(t))\right]. \end{aligned} \tag{17}$$

As a variable approaches a boundary, $\mathcal{E}_{\text{bar}}(x) \to \infty$. The complete homeostatic energy is

$$\mathcal{E}(x) = \mathcal{E}_{\text{met}}(x) + \alpha \mathcal{E}_{\text{bar}}(x). \tag{18}$$

This yields an explicit constrained-game interpretation rather than an ad hoc rate-matching penalty [12].

7 Learning Rule: Three-Factor Plasticity

7.1 Eligibility traces

A canonical neuromodulated STDP model stores local pre–post coincidence in an eligibility trace:

$$\dot{e}_{ij}(t) = -\frac{1}{\tau_e} e_{ij}(t) + \Psi(s_i(t), s_j(t), u_i(t)), \tag{19}$$

where Ψ may implement timing-dependent potentiation and depression. The important property is synapse-local computability [3].

7.2 Learning signals

The third factor $m_i(t)$ may be a global scalar reward or loss broadcast, a layerwise signal, or a structured learning signal that combines local eligibility with broadcast error-like information, as in e-prop [4].

7.3 Core update

For synapse $j \to i$, learning follows crefeq:three-factor. This rule is the primary algorithm rather than an approximation added after the game is defined.

8 Explainability and Credit: Eligibility/Temporal Shapley

8.1 Why subset evaluation is infeasible

Standard neuron-level Shapley attribution evaluates a value $v(S)$ over many neuron subsets S. If $v(S)$ is task accuracy, this requires an infeasible number of forward passes during training.

8.2 Eligibility-Shapley over plasticity drives

Define neuron i's plasticity drive during window n as

$$\Delta W_i^{(n)} := \left\{ \Delta w_{ij}^{(n)} \right\}_j . \tag{20}$$

We then define a tractable surrogate coalition value that approximates the potential improvement contributed by a set of plasticity drives:

$$v^{(n)}(S) \approx \widehat{\Delta\Phi}^{(n)}(S) := \left\langle G^{(n)}, \sum_{i \in S} \Delta W_i^{(n)} \right\rangle, \tag{21}$$

where $G^{(n)}$ is an approximate sensitivity of the potential to weight changes over the window. It may come from a critic, a synthetic gradient, or an e-prop-style learning signal [4].

If the surrogate is additive in ΔW_i, the Shapley value is exact and has the closed form

$$\varphi_i^{(n)} = \left\langle G^{(n)}, \Delta W_i^{(n)} \right\rangle. \tag{22}$$

Thus, marginal contribution can be computed without exponential subset evaluation.

8.3 Shapley–Taylor interactions

To model synergy without combinatorial blowup, one may compute Shapley–Taylor indices through order k, for example $k = 2$, over small circuits or layer blocks rather than over all individual neurons [11].

8.4 Training-time modulation

Eligibility-Shapley credit can allocate neuromodulatory gain according to

$$m_i(t) = m_{\text{global}}(t) \frac{\text{softplus}(\varphi_i^{(n)})}{\sum_{k \in \mathcal{G}} \text{softplus}(\varphi_k^{(n)})}, \tag{23}$$

where $\mathcal{G}$ is a layer or circuit. The normalization discourages every neuron from increasing its activity merely to capture more global credit.

8.5 Post-hoc explanation

The post-hoc interpretation is consistent with both SHAP, which treats features as players [9], and Neuron Shapley, which treats neurons as players [10]. The distinction is that Eligibility-Shapley supplies a tractable training-time surrogate based on plasticity contributions.

9 Backpropagation as a Limiting Case

Backpropagation emerges under three non-biological limiting assumptions:

1. spikes are replaced by smooth rates $r_i(t) \approx \sigma(u_i(t))$;
2. the learning signal is the exact derivative $m_i(t) = -\partial \mathcal{J} / \partial r_i(t)$; and
3. the eligibility trace equals the local Jacobian $e_{ij}(t) \approx \partial r_i(t) / \partial w_{ij}$.

Under these assumptions,

$$\Delta w_{ij} \propto -\int \frac{\partial \mathcal{J}}{\partial r_i(t)} \frac{\partial r_i(t)}{\partial w_{ij}} \, dt = -\frac{\partial \mathcal{J}}{\partial w_{ij}}, \tag{24}$$

which is gradient descent. This is consistent with eligibility-propagation perspectives that relate synapse-local traces and learning signals to gradient learning [4].

10 Induced-Feature Shifts versus Normalization

A natural question is whether Batch Normalization or Layer Normalization could handle the relevant distribution shifts. Normalization layers treat representation drift as a nuisance to be standardized away. In Neuron Games, upstream changes are strategic state transitions: they reshape downstream best-response landscapes and are coordinated by $\Phi(x)$, especially through kernel or DPP diversity, which prevents collapse, and metabolic barriers, which prevent runaway firing. Normalization is therefore a statistical correction, whereas the potential is an explicit coordination mechanism.

11 Practical Windowed Algorithm

For each window $[t_n, t_{n+1}]$, perform the following operations:

1. Roll out the spiking dynamics with $W^{(n)}$ to obtain $S^{(n)}(\cdot)$ and $U^{(n)}(\cdot)$.

2. Update the eligibility traces $E^{(n)}(\cdot)$ locally at every synapse [3].

3. Compute or estimate $m_i(t)$ using global, layerwise, or structured learning signals [4].

4. Update synapses using the three-factor rule in crefeq:three-factor.

5. Update the diversity objective using kernel redundancy or DPP log-determinants on spike or trace objects [5, 8].

6. Apply the metabolic energy and barriers through explicit penalty gradients or projections [12].

7. Optionally compute Eligibility-Shapley credits $\varphi_i^{(n)}$ to allocate modulatory gain and report neuron- or circuit-level explanations [9].

12 Experimental Protocol

12.1 Tasks

The evaluation should include static classification with temporal coding on MNIST and Fashion-MNIST, as well as temporal benchmarks such as spike-based sequence classification, event-based datasets, and toy copy or repeat tasks.

12.2 Baselines

The primary baselines are surrogate-gradient backpropagation, e-prop or related broadcast-learning-signal variants [4], and reward-modulated STDP with a global scalar modulator [3].

12.3 Ablations

The proposed mechanisms should be isolated through the following comparisons:

- **Diversity:** none versus kernel redundancy versus MMD versus DPP log-determinant;
- **Homeostasis:** none versus metabolic cost alone versus barrier alone versus both; and
- **Shapley:** post-hoc attribution alone versus Eligibility-Shapley modulation versus groupwise Shapley–Taylor interactions.

12.4 Explainability metrics

We propose three families of metrics:

- **Faithfulness:** removing the highest-credit neurons or circuits should cause the largest decrease in performance, following Neuron Shapley evaluations [10];
- **Stability:** attribution should remain consistent across data augmentations and adjacent time windows; and
- **Sparsity and structure:** credit concentration in small, interpretable circuits should be quantified when appropriate.

13 Discussion

Three-factor learning, eligibility traces, spike-train kernels, and Shapley explainability are not individually new. The claimable novelty is their unification in a biology-forward game formulation in which

1. induced-feature dependence requires a state-based Markov game;
2. kernel or DPP diversity and metabolic barriers make the game stable and non-degenerate; and
3. Eligibility-Shapley makes game-theoretic credit assignment tractable and allows attribution to influence learning signals.

The method should not be presented as categorically outperforming backpropagation. A defensible claim is that its performance may approach gradient methods when the learning signals approximate the relevant sensitivities, as in the e-prop account, while providing distinct biological plausibility and interpretability mechanisms [4].

14 Conclusion

We presented Neuron Games for SNNs as a constrained state-based potential Markov game whose state explicitly represents induced-feature dependence. Training is driven by three-factor plasticity, with eligibility traces gated by modulatory signals. Stability is enforced through kernel or DPP diversity and metabolic-plus-barrier homeostasis. Finally, Eligibility-Shapley provides a tractable bridge between game-theoretic credit allocation and Shapley-style interpretability, enabling training-time credit shaping without infeasible subset evaluation.

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